HEARTCARE: Web-based Heart Disease Prediction Using Machine Learning



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FINAL APPROVAL

It is certified that I have evaluated the project **"HEARTCARE: Web-based Heart Disease Prediction Using Machine Learning".** The research and its report were found to be sufficient standard to warrant its acceptance to complete the course of studies of bachelor's degree in Computer Science of Govt. Post Graduate Mardan

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A PROJECT SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF THE BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Has been approved.

1._____ **2._____**

NAME: …… **NAME: ……..**

(External Examiner) **(Supervisor)**

Chairman: _____

AUTHORS DECLARATION

The material presented in this thesis, **"HeartCare: Web-Based Heart Disease Prediction Using Machine Learning,"** is entirely original and has been prepared by us for the fulfillment of our Bachelor of Science (BS) degrees at Government Post Graduate College Mardan. We have taken care to acknowledge and thank everyone whose words and concepts we have used. We conducted the research in an honest manner, not attempting to mislead or alter any data. We obtained the necessary approvals and carried out all our actions in a just and proper manner. We all contributed significantly to the idea generation, planning, execution, and comprehension of the project's purpose. We did not steal anything without giving credit to the original author. We're showing this to anyone for the first time today.

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DEDICATION

We dedicate this thesis, "**HeartCare: Web-Based Heart Disease Prediction Using Machine Learning**," to our families and loved ones. Your unwavering support, encouragement, and prayers have been our greatest source of strength throughout this academic journey.

A special thank you to our supervisor, **Mr. Muhammad Assad**, Professor of Computer Science at Government Post Graduate College Mardan. Your guidance, expertise, and mentorship have been invaluable in shaping this work and inspiring us to reach higher.

ABSTRACT

Heart disease is a leading cause of death worldwide, yet early detection remains challenging in many areas. This study presents a web-based heart disease prediction tool that uses machine learning to assess risk from routine health data. We trained three classifiers Random Forest Classifier, Support Vector Machine C (SVC), and XG Boost on the public UCI Heart Disease dataset, which includes features such as age, blood pressure, cholesterol, and chest pain type. In our tests, XGBoost achieved the best performance, with an area under the ROC curve (AUC) of 0.91, while Random Forest reached an AUC of 0.91 and SVC an AUC of 0.89. Based on these results, the XG Boost model was integrated into a Flask-based backend. The frontend, built with HTML, CSS, and JavaScript, offers a responsive form where users enter their data and receive an immediate risk estimate. This non-invasive, low-cost system can help patients and clinicians identify high-risk individuals for timely intervention. Future work will focus on adding more clinical features, securing patient data, and validating the model on larger, real-world datasets. Our study demonstrates how combining a high-performing machine learning model with an accessible web interface can support early heart disease screening and improve patient outcomes.

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Chapter 1. Introduction

1.1 Background of Study

Heart disease or Cardiovascular diseases (CVDs) are the leading cause of death in the world. According to WHO (World Health Organization) every year 17.9 million deaths caused by heart disease and one out of three people who die from heart disease or cardiovascular diseases are under 70 years old [1].

The most dangerous lifestyle that can increase the chances of heart disease are unhealthy diet, less physical activity, smoking or tobacco use and drinking alcohol. The air pollution and other environmental damage. These habits can raise blood pressure, raised blood glucose, raised overweight and obesity.

According to the World Health Organization if a person stop using tobacco, use less salt in the diet, eat healthy food like fruits and vegetables, do physical activity and avoid alcohol can reduce the cause of heart problems [2].

To prevent heart disease, we need to change our lives, stop smoking and alcohol, eat healthy and less salt foods and make exercise our regular routine. Also the government should take responsibility and take actions to make healthy foods cheaper and access to everyone and also build free air-pollution environment.

Heart disease causes two big things deaths and economic crisis, we need urgent actions and strong policies that tackle the root causes of heart disease and also good treatment and community programs to teach people why heart disease happens.

1.1.1 The State of Heart Disease in Pakistan

In Pakistan the heart disease is worse than most the countries in the world. A study 2019 by the Global Burden of Disease reported 357.88 deaths per 100,000 people from Heart disease in Pakistan and globally 239.85 per 100,000 [3]. In Pakistan most people

common risk factor in population are high blood pressure, diabetes, smoking and obesity. Because many people have these risks, which may cause heart disease problem in large population of the country.

1.1.2 Limitations of Conventional Diagnostic Methods

The doctors use exams and tests like physical checkups and ECG to predict the heart disease. These methods are costly many people cannot afford it, in some parts of the country does not have access to these tools. Because of this, many people do not know they are high risk of heart disease until they have a serious incident.

1.1.3 Machine Learning for Improved Prediction

Machine learning is a powerful tool to use for heart disease prediction. Machine learning find hidden patterns in data for better prediction. Machine learning model looks at patient data like their age, blood pressure, cholesterol and other factors to predict the heart disease. Machine learning can be inexpensive than tests like ECG or angiograms.

The goal of this study is to build a machine learning model uses common health data to predict heart disease, which can prevent serious illness and reduce the high number of cardiovascular diseases deaths.

1.2 Problem Statement

Heart disease kills millions people annually, many people suffer long-term heart disease problems even if they survive against a heart attack or stroke. The available methods like Electrocardiograms are expensive most people cannot afford this test and also the Electrocardiograms may not capture the irregularities if patient heart is normal at test time. In many parts of Pakistan, Clinics not have ECG machines and other lab facilities. The poor urban areas cannot afford specialist care. Without affordable and easy-to-use

tools, large parts of population remain undiagnosed. Because of undiagnosed many patients miss early treatment and continue suffer heart problems.

1.3 Objectives of the Study

To solve the problem of unaffordable and late heart disease diagnosis this study focus create tool that easy to access and affordable:

1. **Develop and Train Machine learning Model:**

We will use the publicly available UCI Heart Disease Dataset [4] to train different machine learning models like random forest Classifier, XG Boost Classifier, and Support Vector Machine Classifier. This dataset contain 920 records of four different places Cleveland, Switzerland, Hungary and VA Long Beach. This dataset is useful for training models for the binary classification problems.

2. **Compare Model Performance:**

After train the all three machine learning models, we will evaluate the performance of the three models using common evaluation metrics like accuracy, precision, f1-score and recall. The goal is to identify which model is better than other models.

3. **Deploy the best Model:**

After successfully train model and identify the best model, we will create a simple, interactive web-based interface. Where a user can enter their health information and model will predict that the user have heart disease or not.

The main aim is to build a practical, low-cost and easy-to-use tool that can help to detect heart disease early.

1.4 Scope and Limitation

The scope of this study is limited to the following areas:

1. **Machine Learning Models:**

The research focused on three machine learning models Random forest Classifier, XG Boost and Support Vector Machine Classifier. These model are great to start in machine learning and perform well on small datasets.

2. **Data:**

The data we are using to train machine learning models on available UCI dataset, which are publicly available. This dataset contain 13 clinical features which help to predict heart disease. We will use this data to train, test and validating using this dataset.

3. **Final Step**

The main outcome of this project is a working web-based tool. This website will get data from user using form and send it to the model for prediction and show predicted class (heart disease or not heart disease) to the user.

1.4.2 Limitations

It is important to acknowledge the limitations of this study:

4. **Limited Dataset**

The machine learning models are only trained on UCI Heart Disease dataset, which contain 920 patients' records. This dataset may not represent all countries age groups or health background. Also this dataset only include 13 features.

5. **Only Three Models Trained**

The study focused on only three machine learning models Random Forest Classifier, XG Boost Classifier and Support Vector Machine Classifier (SVC). Other advanced models like neural networks are not used. May be other models perform better then trained ones.

6. **Not a Medical Diagnostic Tool**

The web-based tool are only prototype which only the idea of predicting heart disease risk. This tool cannot replace a doctor because it not an official diagnostic tool.

7. **Basic Website**

The website which we create for this project does not include advanced features like user login or authentication, secure data protection and integration with hospital systems.

Chapter 2. Literature Review

The researchers use various techniques for heart disease prediction using machine learning. They use different algorithms like Decision Tree, Random Forest Classifier and advanced techniques like neural networks and deep learning to train the models. The research show us how helpful machine learning can be healthcare.

2.1 Heart Disease and Its Risk Factors

Heart disease is the major cause of death worldwide, making one of the biggest challenge for global health. According to the World Health Organization (WHO) heart disease problems like coronary artery disease, heart failure and stroke are remain top concern in world health. These diseases are caused by a mix of genetic, unhealthy food, less physical activity and environmental factors.

Factors such as genetics, environment, or other influences may play a role in the development of heart disease. This includes conditions like high cholesterol, elevated tobacco usage, Type 2 diabetes, kidney disorders, and certain lifestyle choices, including smoking, insufficient physical activity, unhealthy eating habits, and excessive alcohol consumption. Genetic predispositions or complications during pregnancy, such as hypertension and gestational diabetes, can also increase the likelihood of developing heart disease [5].

2.2.1 Types of Risk Factors

The risk factors of heart disease are mainly divided into 2 types.

2.2.1.1 Non-Modifiable Risk Factors

Non-modifiable are those factors that cannot be changed like age as people get older the risk of heart disease increases, sex man are at higher risk than woman before menopause and family history a person close relative (parent or sibling) who had heart disease at early age has higher chance to developing themselves. These factors are important to identify people with high risk heart disease.

2.2.1.2 Modifiable Risk Factors

Modifiable risk factors are related to lifestyle or medical factors that can be changed, treated and control like high blood pressure, high cholesterol (LDL), diabetes mellitus, smoking and tobacco use, obesity and lack of physical activity.

The INTEHEART Study by Yusuf et al. [6], which looked at heart attacks around the world and found that nine modifiable risk factors were responsible for over 90% of all heart attacks. This include high blood pressure, high cholesterol, diabetes, stress, obesity, lack of exercise and smoking.

2.2 Machine Learning in Medical Diagnosis

In the recent years, machine learning become one of the powerful and important tool in many fields including healthcare and medical diagnosis. As we know a computer can process large amount of medical data much faster and accurate. The machine learning algorithms and analyze complex data that would be impossible for a human to study it manually. A machine learning model can find hidden patterns in multiple factor data. For example a machine learning model can predict heart disease by analyzing different factors like age, chest pain, blood pressure, high cholesterol, and diabetes and smoking

habits. The machine learning algorithms can learn from data and make accurate prediction about whether a person have heart disease or not.

2.3 Use of Machine Learning in Heart Disease Prediction

In cardiology field, the machine learning already being used to improve the accuracy and speed of heart disease prediction. Traditional methods like ECG or physical checkups have limitation like time consuming and requiring specialist. But a machine learning model can quickly analyze clinical factors and give prediction.

Many research studies have use different machine learning algorithms and found which one perform well and better choice than other.

El-Sofany [7] (2024) conducted a detailed investigation into predicting heart disease through various machine learning algorithms and data classification methods. This research was published in IEEE Access, which underscores its peer-reviewed reliability. The study employed classification techniques like Support Vector Machines (SVC), Random Forest, and additional ML models, while implementing preprocessing methods such as SMOTE to manage class imbalance. The effectiveness of each algorithm was assessed using several accuracy metrics. Findings indicated that some machine learning models achieved higher prediction accuracy than others. The research also highlighted the importance of explain ability by utilizing SHAP (Shapley Additive explanations), aiding in the interpretation of model decisions. This study underscores the significance of selecting appropriate models and understanding feature importance within clinical diagnosis systems. However, the study did not make the model available in a user-friendly format, a gap that our research seeks to fill with a web-based heart disease prediction application.

Fitriyani et al. [8] (2020) created an enhanced Heart Disease Prediction Model (HDPM) intended for use in a Clinical Decision Support System (CDSS). This model integrated three key methodologies: DBSCAN for identifying and eliminating outliers, SMOTE-ENN for addressing class imbalance within the dataset, and XG Boost as the ultimate classification algorithm. The research utilized two well-known heart disease datasets — Stat log and Cleveland — and evaluated the proposed model's effectiveness in comparison to traditional machine learning algorithms like Naïve Bayes, Logistic Regression, SVC, Decision Tree, Random Forest, and MLP. Their model reached an accuracy of 95.90% on the Stat log dataset and 98.40% on the Cleveland dataset, surpassing the performance of existing models. Furthermore, they created a prototype CDSS interface to support physicians in the early diagnosis of heart disease based on patient information. This research highlights the effectiveness of hybrid preprocessing and ensemble learning, although it emphasizes accuracy more than integration with contemporary web-based interfaces, which our research aims to address.

Almazroi et al. [9] (2023) proposed a deep learning-based Clinical Decision Support System (CDSS) for heart disease prediction using dense neural networks implemented in Keras. The study focused on analyzing multiple neural network architectures by varying the number of hidden layers (from 3 to 9), each containing 100 neurons and using ReLU activation functions. The model was tested on multiple heart disease datasets and evaluated using key metrics such as accuracy, sensitivity, specificity, and F-measure. In addition to comparing various deep learning configurations, the authors also tested ensemble approaches. The results showed that their deep learning model outperformed individual machine learning classifiers and other ensemble models across

all datasets. The study concluded that deep learning is highly effective for accurate heart disease prediction. Although the work focused on structured data, the authors suggested future improvements by integrating image data through Convolutional Neural Networks (CNNs), which would enable automatic feature extraction and potentially further enhance predictive performance. This research reinforces the growing role of deep learning in medical diagnostics and supports our study's emphasis on exploring advanced algorithms for real-time heart disease risk prediction.

Reshan et al. [10] (2023) put forward a comprehensive heart disease prediction system utilizing Hybrid Deep Neural Networks (HDNN) that integrates various deep learning models like Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM). The research presented a CNN-LSTM hybrid model augmented with additional dense layers to enhance diagnostic precision. They assessed the system using two datasets: the Cleveland Heart Disease dataset and a substantial combined dataset (including Switzerland, Cleveland, Stat log, Hungarian, and Long Beach VA). The suggested CNN-LSTM model demonstrated remarkable outcomes, achieving 97.75% accuracy on the Cleveland dataset and 98.86% accuracy on the larger dataset, surpassing traditional machine learning and existing deep learning models in terms of accuracy, sensitivity, specificity, precision, AUC, F1-score, and MCC. The system also incorporated data imputation and feature selection techniques such as the Extra Trees Classifier to improve the quality of model inputs. This research illustrates the promise of hybrid deep learning architectures for the precise and dependable clinical prediction of heart disease and is in line with the ongoing research trend of employing advanced neural network methods for enhanced healthcare decision support.

Qadri et al. [11] (2023) introduced a novel feature engineering approach known as Principal Component Heart Failure (PCHF) aimed at the early prediction of heart failure through machine learning. The research utilized a dataset containing 1025 patient records and tested nine machine learning algorithms, including Decision Tree (DT), Logistic Regression, Random Forest, SVM, Naive Bayes, KNN, XG Boost, Gradient Boosting, and MLP. The PCHF technique successfully identified the eight most significant features to create an optimized dataset, thereby enhancing prediction efficiency and accuracy. Of all the models analyzed, the Decision Tree yielded the highest performance, attaining 100% accuracy with a minimal runtime of just 0.005 seconds. The researchers also employed 10-fold cross-validation to verify their findings and established that their model surpassed the results of previous studies. This research underscores the significance of sophisticated feature engineering, illustrating how the judicious selection of features can greatly improve heart disease detection. Although the emphasis was specifically on heart failure, rather than heart disease in general, the method of utilizing feature engineering alongside various machine learning models closely resembles the approach taken in our study.

Ahmad et al. [12] (2022) the study investigated the impact of hyperparameters tuning on heart disease prediction by assessing four widely used machine-learning algorithms—Logistic Regression (LR), k-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Extreme Gradient Boosting Classifier (XG Boost)—with and without the application of Grid Search CV. They employed a 5-fold cross-validation method to train and evaluate their models across various public datasets (Cleveland, Hungary, Switzerland, Long Beach VA, and UCI Kaggle). The results revealed that

XG Boost with Grid Search CV achieved the best performance, reaching up to 100% training accuracy and 99.03% testing accuracy, while XG Boost without tuning still produced notable results (100% train, 98.05% test). By systematically adjusting the parameters, the authors demonstrated the potential of hyperparameters optimization to improve predictive performance. However, the research mainly focused on accuracy measures and did not explore advanced ensemble techniques or real-world applications, which our study aims to tackle by integrating multiple models (Random Forest, XG Boost and SVM) and creating an optimal model as a web-based tool for healthcare professionals.

Shamout et al. [13] (2021) offer an extensive overview of current machine learning (ML) applications aimed at forecasting clinical outcomes through electronic health records (EHRs). These predictive models are created to aid clinical decision-making and have been utilized for various conditions, such as mortality, cardiac arrest, acute kidney injury, and arrhythmia. The review emphasizes the complete process of developing these models, which includes data processing, feature extraction, inference, and model assessment. It also points out significant challenges like modeling assumptions, data quality, and interpretability, while proposing avenues for future research. This study highlights the increasing significance of ML in the healthcare sector and its capability to create efficient clinical decision support systems leveraging big data from EHRs.

Chaudhuri & Das 2024 [14] employed a hybrid approach to create a CVD prediction model, integrating support vector machines, random forests, and Adaptive Boosting algorithms. The SVC Attribute Assessment technique was applied to identify important

variables, whereas random forests were used to choose instances with high weights. This method surpassed both individual classifiers and ensembles of classifiers in predicting CVD, even with a limited set of features, showcasing its effectiveness.

Saleem et al. [15] (2025) the Neural-XG Boost (N-XGB) model has been introduced as a hybrid approach that merges neural networks (NN) for in-depth feature extraction with XG Boost for accurate classification. This model addresses a multi-class classification problem related to disasters such as wildfires, floods, and earthquakes. By employing SMOTE to tackle class imbalance and combining the strengths of deep learning with gradient boosting, the N-XGB model achieved an impressive accuracy of 94.8% and an average F1-score of 0.95, significantly outperforming traditional models like Random Forest, SVC, and Logistic Regression. Its consistent performance across all types of disasters (wildfires: 0.96, floods: 0.93, earthquakes: 0.96) highlights its dependability. The study also emphasizes the importance of hyperparameters tuning and data preprocessing in obtaining optimal performance. Future research will aim to evaluate the model on a broader set of datasets.

Chapter 3. Methodology

This chapter outlines the methodology used for predicting heart disease through Random Forest Classifier, Extreme Gradient Boost and Support Vector Machine Classifier algorithms. It describes the data collection process, preprocessing steps, model development, and model evaluation techniques.

3.1 Data Source

We use a publicly available dataset known as UCI Heart Disease Dataset to train and test our machine learning models. This contains 920 records from four different locations. Each record includes 13 input features and 1 target variable (num). The details of this data are provided in Table 1.

Feature Name	Description	Range
age	Age of the patient in years	28 to 77
dataset	Place of study	Cleveland, Hungary, Switzerland, VA Long Beach
sex	Male/Female	Male & Female
cp	Type of Chest Pain	1. typical angina 2. atypical angina 3. non-angina 4. asymptomatic
trestbps	resting blood pressure (in mm Hg on admission to the hospital)	0 to 200
chol	serum cholesterol in mg/dl	0 to 603
fbs	if fasting blood sugar > 120 mg/dl	False (692) True (138)

		Missing (90)
restecg	resting electrocardiographic results	normal (551) lv hypertrophy (188) st-t abnormality (179) Missing (2)
thalch	maximum heart rate achieved	60 to 202
exang	exercise-induced angina (True/ False)	False (528) True (337) Missing (55)
oldpeak	ST depression induced by exercise relative to rest	-2 to 6
slope	the slope of the peak exercise ST segment	flat (345) upsloping (203) downsloping (63) Missing (309)
ca	number of major vessels (0-3) colored by fluoroscopy	0,1,2,3
thal	[normal; fixed defect; reversible defect]	normal (196) reversible defect (192) fixed defect (46) Missing (486)
num	the predicted attribute	0,1

3.1 1 Features

Note: Missing indicates records where feature was not recorded.

This table summarizes all clinical features (excluding the dataset feature) used to train and test the machine learning models.

3.2 Data Exploration

Data exploration is very important to understand the data correctly. We explored our dataset to understand the distribution of our features, identify the outliers and the relationship between input features and target variable. This step help us to uncover the hidden patterns in our data.

3.2.1 Numerical Data Distribution

We visualized the numerical features like age, chol and trestbps using histogram. Histograms show the distribution of our data.3

3.2.1.1 Age Distribution

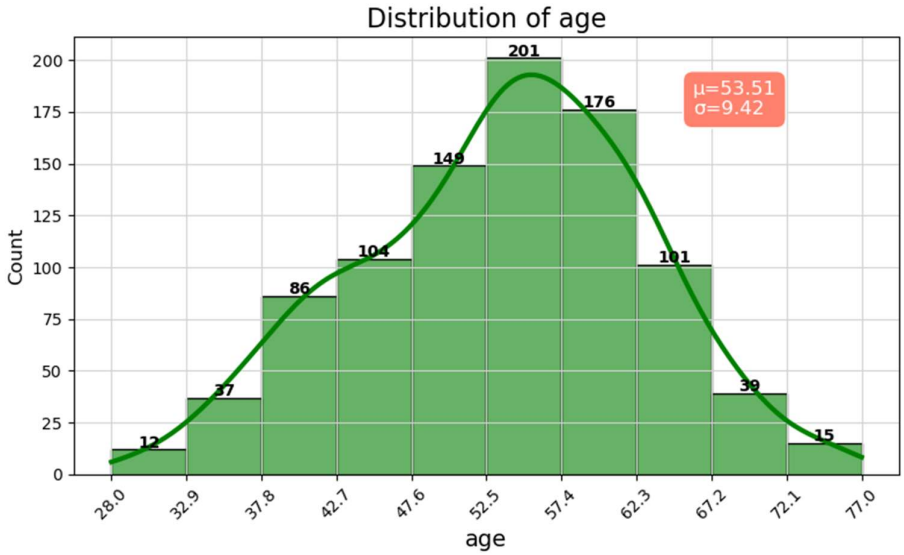


Figure 3.2.1 1 Age Distribution

From histogram we can clearly see that most patients are in range 52 to 62. There are also fewer younger and older on our data.

3.2.1.2 Trestbps Distribution

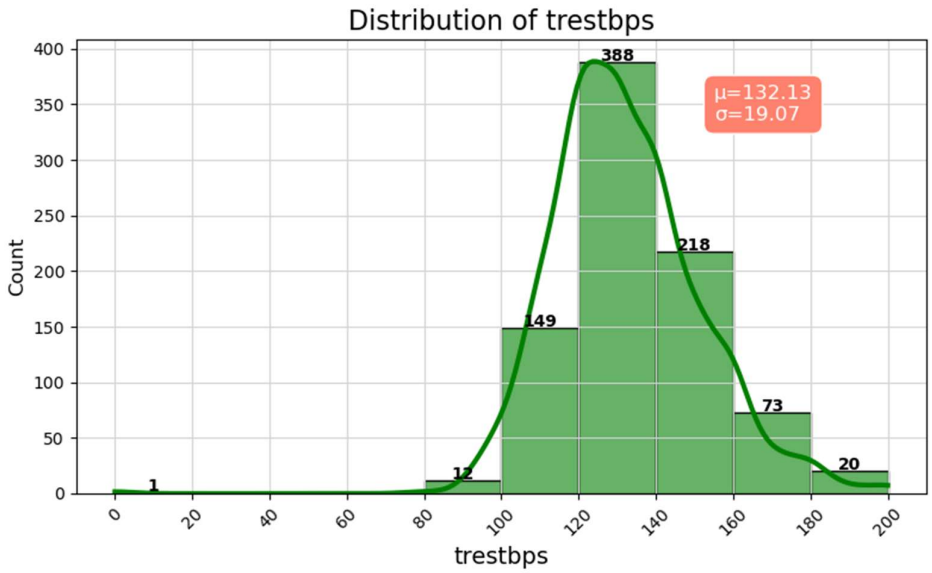


Figure 3.2.1 trestbps Distribution

Most people cluster between 120 and 140, a common normal-to-high range and a long tail toward higher pressures shows some with very high blood pressure.

3.2.1.3 Chol Distribution

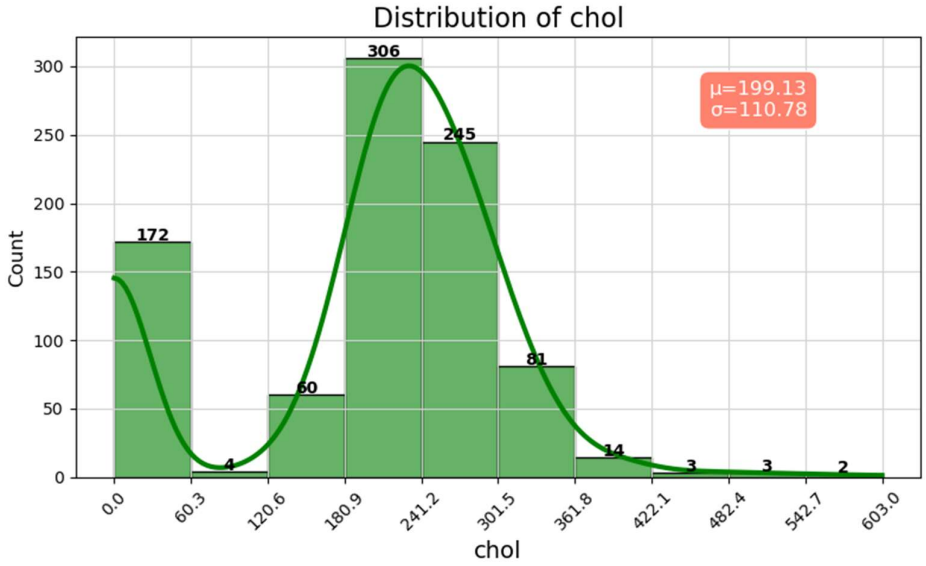


Figure 3.2.1 2 chol Distribution

Cholesterol ranges widely from 126 to over 500 mg/dl. Most patient's cholesterol levels lies between 180 to 300, there is also 172 values of cholesterol from 0 to 60,

3.2.1.4 Thalch Distribution

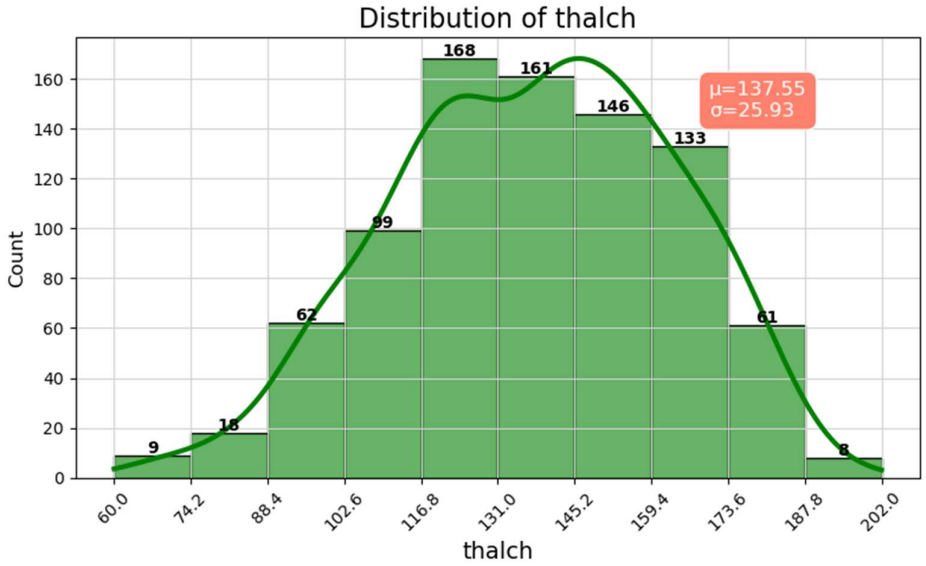


Figure 3.2.1 3 thalch Distribution

Thalch range from 60 to about 202 bpm. Most are between 116 and 160 beats per minute. The bell-shaped curve suggests a fairly normal spread around the mean.

3.2.1.5 Oldpeak Distribution

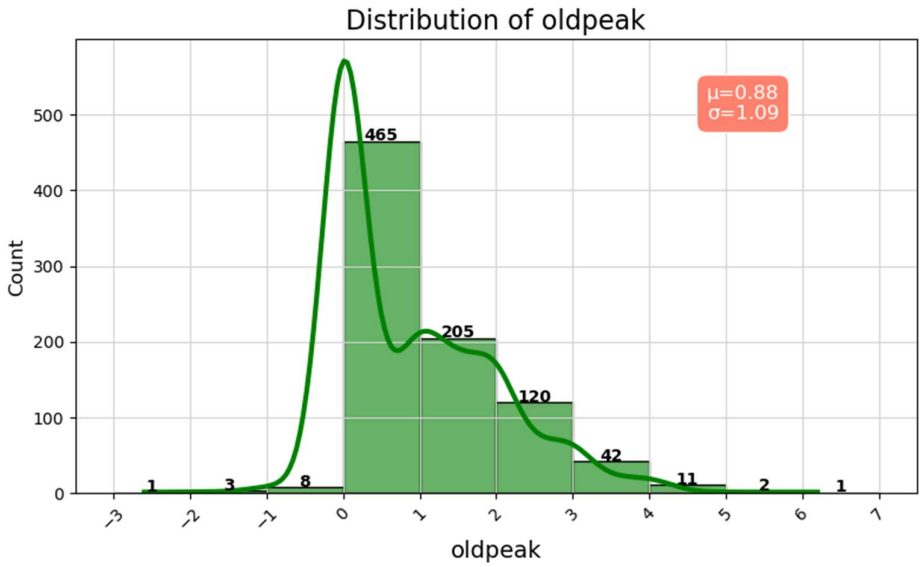


Figure 3.2.1 4 oldpeak Distribution

Oldpeak Column Most values lie between 0 to 3, and few values are >3 or <0

3.2.1.6 ca Distribution

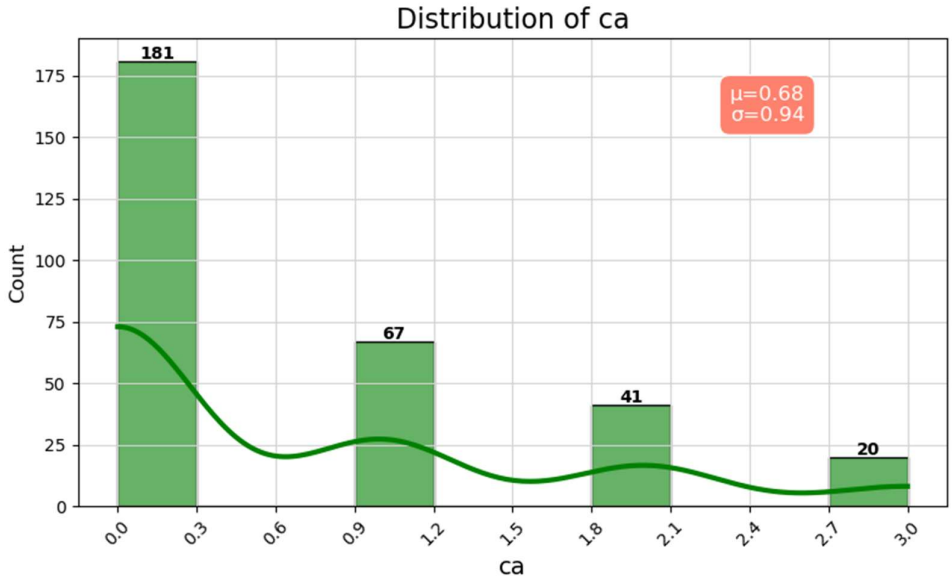


Figure 3.2.1 5 ca Distribution

Most patients have 0 vessels (about 180 cases). Fewer have 1 or 2, and very few have 3.

3.2.1.7 Num Distribution

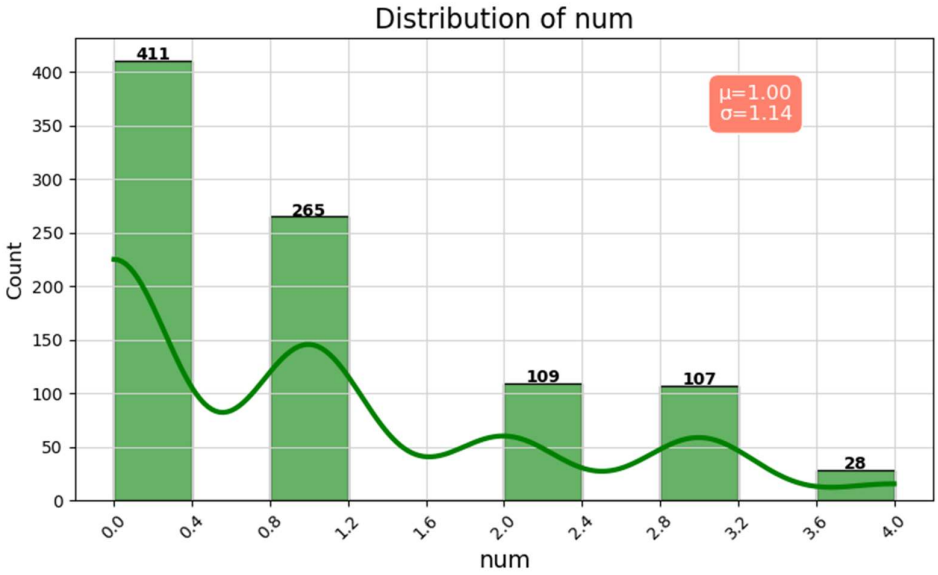


Figure 3.2.1 6 Target Column Distribution

In our dataset there are 411 no heart disease record and from 1 to 4 total 509 patients have heart disease.

3.2.2 Categorical Data Distributions

We visualized the categorical that using bar chart to understand it more accurately.

3.2.2.1 Sex Distribution

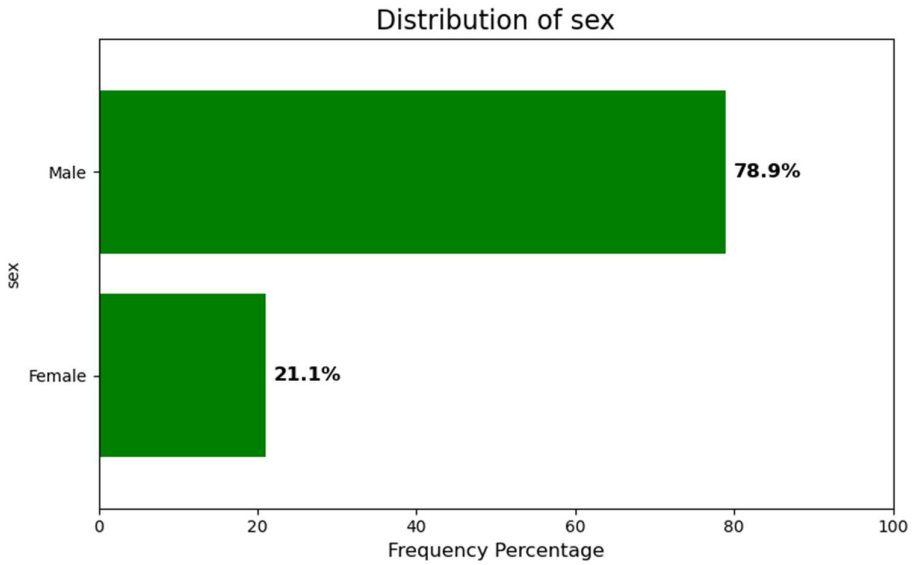


Figure 3.2.2 1 Sex Distribution

In dataset male are 79% and female are 21%. There are almost four times as many men as women in this dataset.

3.2.2.2 CP Distribution

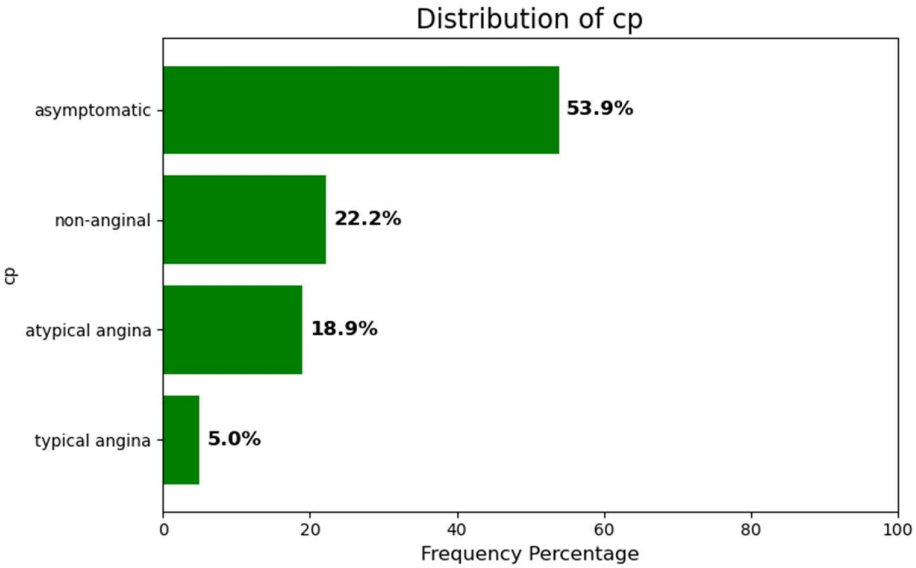


Figure 3.2.2 2 cp Distribution

In dataset Asymptomatic 54%, Non-angina 22%, atypical angina 19, typical angina 5%.

3.2.2.3 FBS Distribution

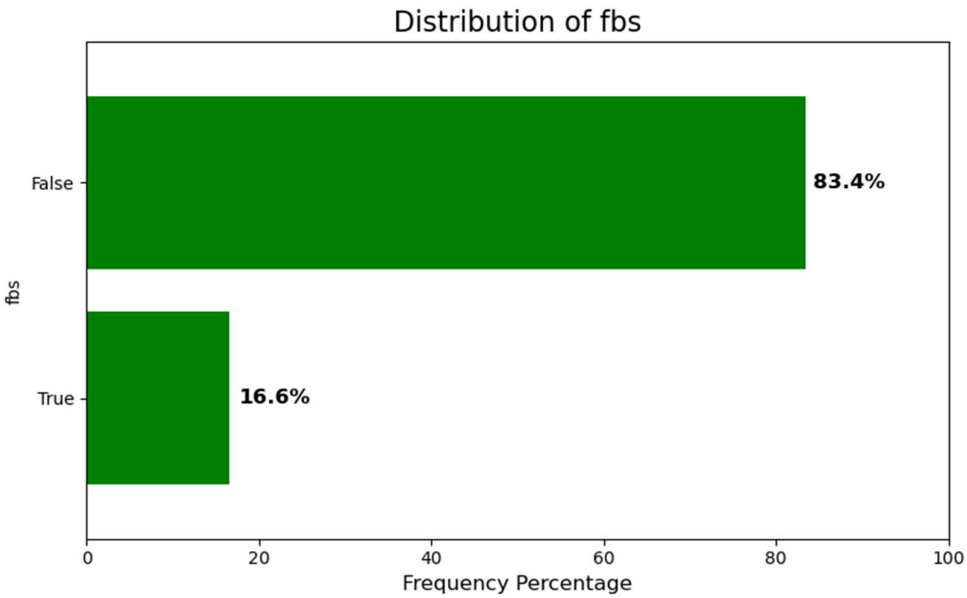


Figure 3.2.2 3 fbs Distribution

In our data False 83%: most people did not have high fasting blood sugar and True 17%: a minority had elevated fasting sugar.

3.2.2.4 Restecg Distribution

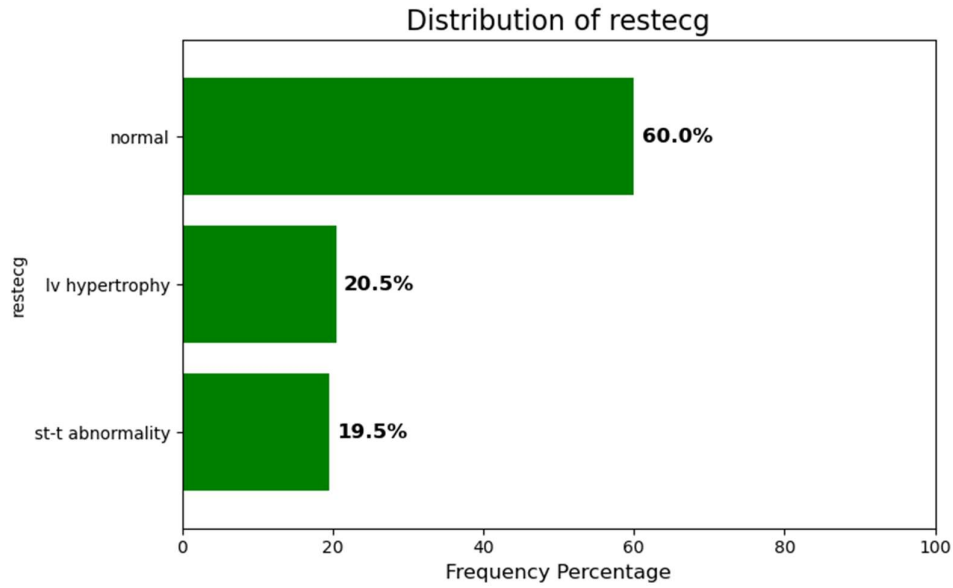


Figure 3.2.2 4 restecg Distribution

Normal 60%: Left-ventricular hypertrophy 20%: ST-T abnormality 20%.

3.2.2.5 Exang Distribution

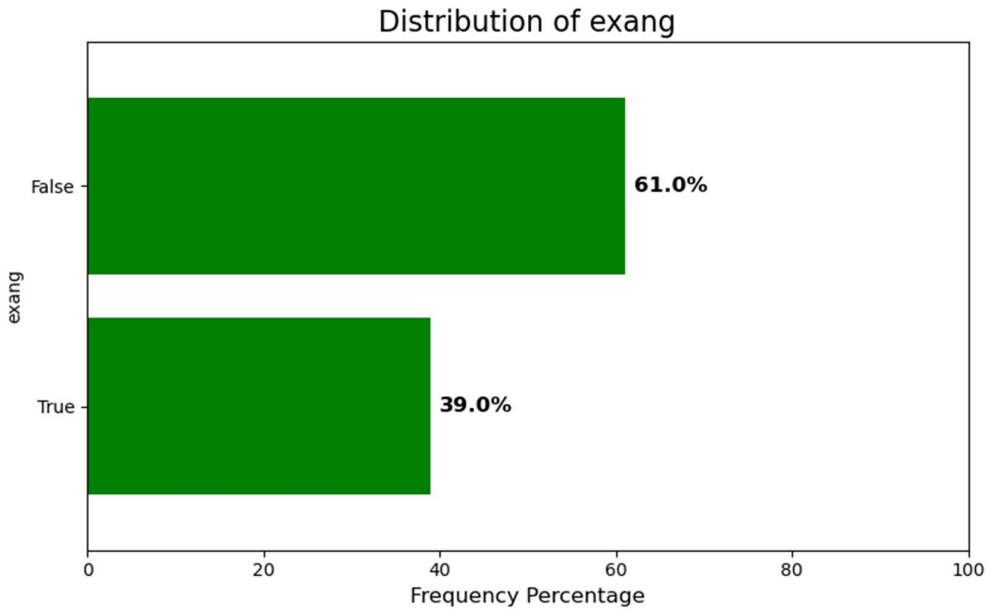


Figure 3.2.2 5 exang Distribution

False 61%: did not chest pain during exercise. True 39%: have chest pain when active or exercising.

3.2.2.6 Slope Distribution

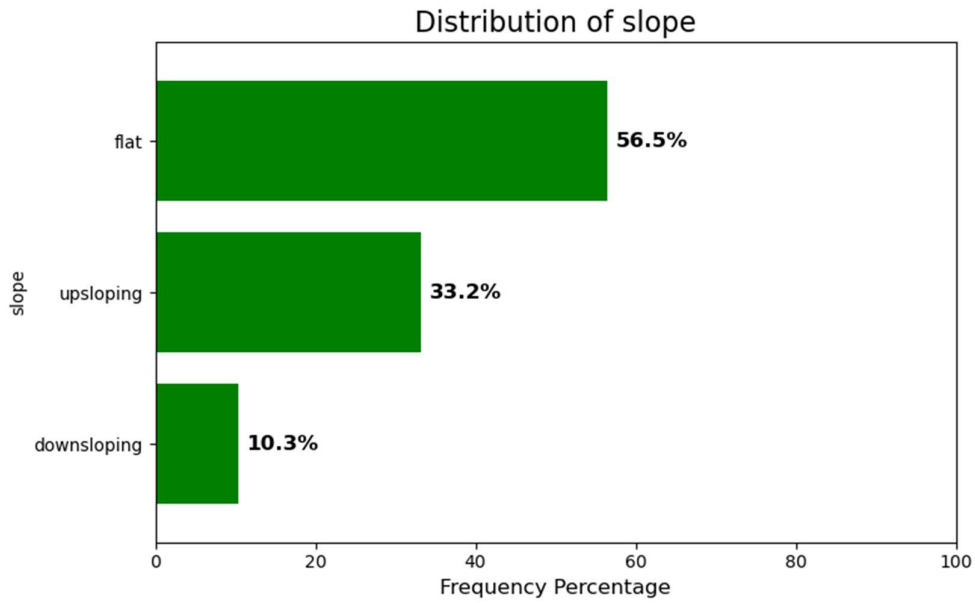


Figure 3.2.2 6 slop Distribution

3.2.2.7 Thal Distribution

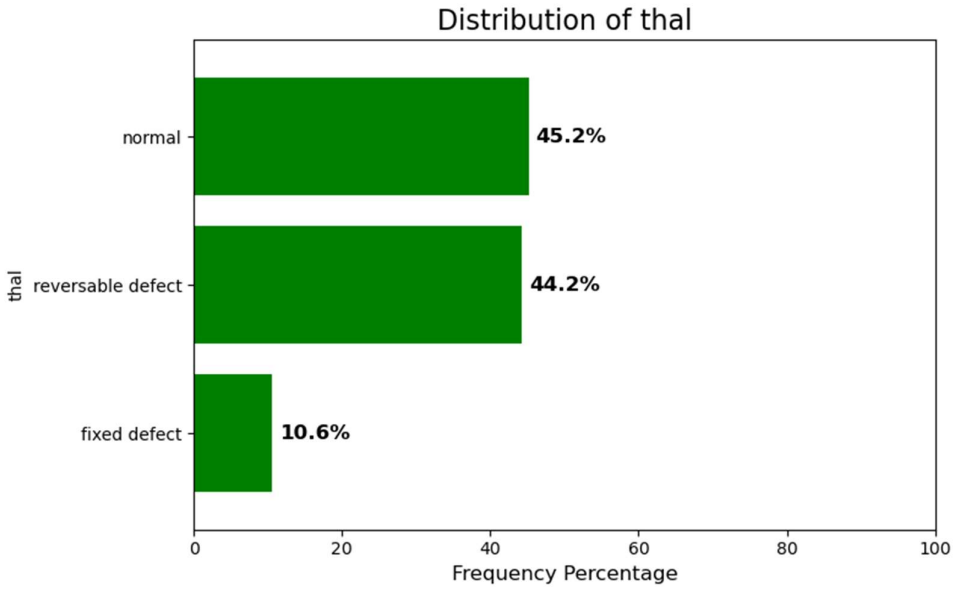


Figure 3.2.2 7 thal Distribution

Normal 45%, Reversible defect 44%, fixed defect 11

3.3 Data Preprocessing

3.3.1 Handling Missing Data

To training any model, it is important to address any missing values within the dataset. Incomplete data can skew our findings and reduce the model's accuracy. The initial examination of the Heart Disease UCI dataset showed that various features have a considerable amount of missing values. Below is a summary of the total missing values in each feature:

Variable	Count of Missing Values
fbs	90
restecg	2
exang	55
slope	309
thal	486
trestbps	59
Chol	30
thalch	55
oldpeak	62
Ca	611

3.2.1 1 Missing Values

To tackle the considerable presence of missing values found in the dataset, we will implement an advance model-based imputation technique to maintain the integrity and predictive ability of our data. We can use basic statistical methods such as mean or

median imputation, but it can change the natural distribution and relationships within the data, we use a model-based imputation approach which views each feature with missing values as a target variable and estimates its value using all other features in the dataset. The foundation of our imputation process will involve two powerful ensemble models: the Random Forest Regression for imputing missing numerical features like 'trestbps' and 'oldpeak', and the Random Forest Classifier for categorical features such as 'thal' and 'slope'. By modeling each feature's missing values with its own Random Forest, this approach allows us to capture the complex, non-linear relationships between variables, leading to a more precise and thoroughly completed dataset that is prepared for the final modeling stage.

3.3.2 Handling Outliers

One of the important step in data preprocessing procedure is identifying and finding the outliers. Outliers are these unusual data points can have a significant impact on model training and can reduce the accuracy of the model. To graphically represent outliers in our dataset, we applied boxplots, it is a best way of visualizing the outliers in data distributions. After examining the boxplots for all numerical features, we found that the 'trestbps' (resting blood pressure) column had one distinct outlier. While other features may have shown some data points that are outside the interquartile range, they were thought possibly thoughtful of rare but clinically important conditions. We will provide outliers free data to the model so model accurately learn from data. We decide to eliminate that one outlier.

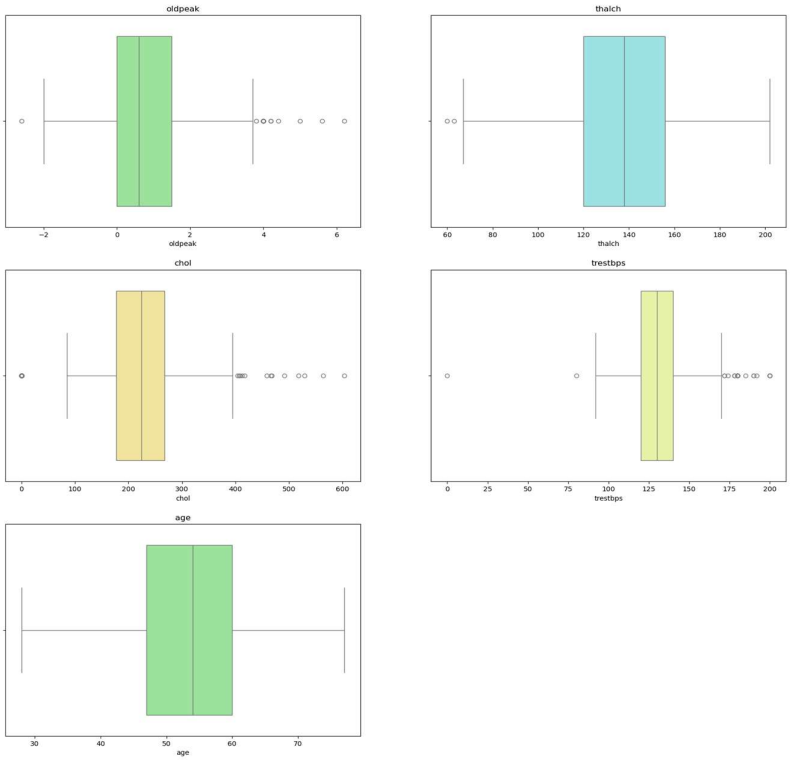


Figure 3.1 1Visualizing Outliers

3.3.3 Converting Categorical to Numerical

In machine learning, some algorithms can only be trained on numerical data. The models we will train are need numerical data. In our data there are categorical data which need to be converted into numerical before give it to algorithm. For conversion we used Label Encoder technique, which converts each categorical value to a unique value.

Like it will convert male and female to 1 and 0.

3.3.4 Feature Scaling

The final step in data preprocessing is feature scaling. The feature scaling makes all numerical input features scaled in same range. In our dataset some feature have large size values like chol (0 to 600) and some have small values like slope (1, 2, and 3).

Some machine learning algorithms rely on distance or gradient calculation like Support Vector Machine, they give more weight to the higher values which can damage the overall performance of the model. To avoid this problem we use minMaxScaler which rescale all numerical data in 0 to 1 range.

3.3.3.1 minMaxScaler Formula

$$Xnew = \frac{Xi - \min(X)}{\max(X) - \min(X)}$$

MinMaxScaler subtract the minimum value for each feature column then divide the result by the original range of the feature (max - min). The new value can be calculated as old value – min value divide by max – min.

The minMaxScaler maintains the shape of each feature distribution. It simple and effective for most machine learning models. After applying minMaxScaler all features lie between 0 to 1. It will help models to learn from data more accurately.

3.4 Model Selection & Training

After the preprocessing steps, we choose three classification algorithms to train on our dataset. The Random Forest Classifier (RFC), Extreme Gradient Boost Classifier (XG Boost) and Support Vector Machine Classifier (SVC) algorithms are trained on our data. And after training we compare their performance to identify which model is better choice of heart disease prediction.

3.4.1 Random Forest Classifier

The first model we trained on data was random forest classifier. Random Forest Classifier creates decision trees, each tree learn from train data and then combine their predictions to achieve more accuracy.

We tuned the parameters of Random Forest using grid search cross validation method to achieve best setting for our model and get high accuracy.

3.4.2 Extreme Gradient Boost Classifier

The second model we trained was the XG Boost which stands for extreme gradient boost. XG Boost is very powerful algorithm, known for its speed and performance in machine learning challenges. The XG Boost algorithm builds sequence of decision trees, and each new tree correct errors made by the last tree. Every new tree in XG Boost try to improve the prediction of previous tree.

For XG Boost algorithm we also apply grid search cross validation method to get better result.

3.4.3 Support Vector Machine Classifier (SVC)

The last model we trained was Support Vector Machine Classifier (SVC). The SVC is one of the powerful algorithm for binary classification problems like ours. The SVC finds the best boundary line also known as hyperplane that separate the data into different classes. We use Radial Basis Function kernel to handles non-linear relationships effectively.

For SVC we also apply the Grid search CV method to get better combination of parameters.

3.4.4 Grid Search CV

Grid Search CV (cross validation) is a technique in machine learning that use for tuning a model hyperparameters, hyperparameters are the variables of a model which control the learning process and performance of model. It try to find the best combination of hyperparameters for a model by using the defined grid. Grid search CV use cross validation technique to access performance of each hyperparameters combination. The grid search CV help us to reduce overfitting and find the best setting for our model.

With the help of grid search CV technique we improved overall performance of the models and will help us to find the best model for our heart disease prediction.

3.5 Web Application Integration

To make easy for user, we developed fully functional web-based application. The user can easily enter their heart related data through a form and they will receive the predicting class on same page. We use HTML, CSS and JavaScript for frontend designing and a python framework called Flask for backend development.

Chapter 4. Web-Based Development

Our project is a web-based, we develop a responsive and interactive website using HTML, CSS and JavaScript, where user can easily enter their health related data. We use a Python framework Flask to handle the backend logic and deploy the XG Boost model in our website, it will predict that the user have heart disease or not. The prediction result will be shown to the user.

4.1 System Architecture

The heart disease prediction system is a client-server design. We create an interactive and responsive (mobile users can also use our website) website. Where user enter their health data in form and the best model (XG Boost) will predict their heart class immediately and show it to the user.

4.1.1 Main Components

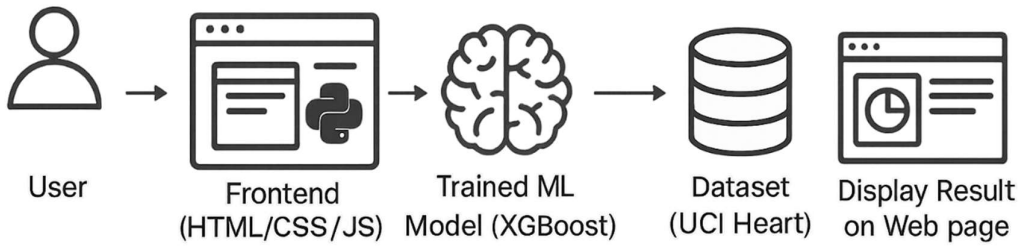


Figure 4.1.1 1 System Architecture

4.1.1.1 Build Interface

To make our tool accessible to everyone and easy to use, we designed and develop a clean, responsive and interactive a user interface using HTML, CSS and JavaScript.

The main goal was to build an interface that simple to use.

After the complete structure of website using HTML, we use CSS to make it interactive and responsive.

We use JavaScript to handle the user inputs and send user data to backend for model prediction. The JavaScript will also receive the response from backend and show it to user with the help of HTML tags.

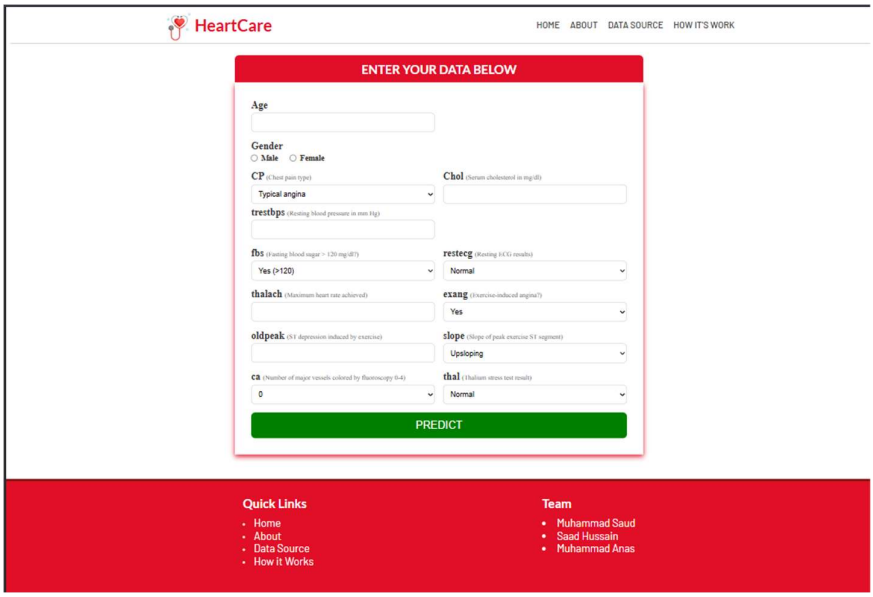


Figure 4.1.1 2 Frontend Interface

4.1.1.2 Backend Development

We use python framework Flask to develop the backend. We create routes to make it easy for user to visit other pages. The Flask app receives data from frontend, apply preprocess steps, load the XG Boost model to predict the class based on user data and send the predicted class back to frontend.

4.1.1.3 Machine Learning Model

We save the XG Boost model using joblib library because of their best performance, and load in flask app with the help of joblib.

- When a user enter their data, it will be sent to the Flask app using JavaScript AJAX.
- In backend the preprocess steps which we applied during training, will apply on user data, then the XG Boost model will predict the class.
- The result is shown to the user on the same page with the message: “Heart Disease” or “No Heart Disease”.

Chapter 5. Result & Discussion

In this chapter, we will see the results that we obtain from training and testing models: Random Forest Classifier, Extreme Gradient Boost and Support Vector Classifier. We train the model using preprocessed data of UCI Heart Disease Dataset.

We evaluated the performance of each model using standard classification report and AUC (Area under the curve). Their result help us to identify best model. After presenting the result, a detailed discussion is provided to explain the performance of each model and how a best model can help us in healthcare applications.

5.1 Model Performance

The performance of all three models are not bad, we train these models on same dataset UCI Heart Disease after proper data preprocessing.

To compare the models we calculate the classification report which show us Precision, Recall, F1-Score, Support and overall accuracy.

a) Accuracy:

Accuracy tells us how many predictions a model correctly guesses in given sample data.

To calculate accuracy we use the following formula:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

b) Precision:

Precision shows us how many model predict correctly positive predictions.

To calculate the precision we use

$$Precision = \frac{TP}{TP + FP}$$

TP: True Positive, model correctly predict positive cases.

FP: False Positive, model incorrectly predict the positive cases.

c) Recall:

Recall shows us how many positive cases are correctly identified by a model.

The Recall Formula is:

$$Recall = \frac{TP}{(TP + FN)}$$

FN: False Negative, model incorrectly predict negative cases.

d) F1-Score:

F1-Score show us the harmonic mean of precision and recall.

$$F1 = \frac{2 * (Precision * Recall)}{(Precision + Recall)}$$

c) Support

Supports show us examples of each class in our dataset, it help us to know data our dataset is balanced or not.

5.1.1 Random Forest Classifier Classification Report

We ran grid search to tuned the hyperparameters

For random forest the best optimal setting by grid search:

8. estimators: 150 (number of trees)
9. min_samples_split: 10
10. min_samples_leaf: 1
11. max_depth: None (no limit)

With these settings Random Forest Classifier achieved:

	PRECISION	RECALL	F1-SCORE	SUPPORT
0 (No Disease)	0.80	0.81	0.81	123
1 (Disease)	0.85	0.84	0.84	153
accuracy			0.83	276
macro avg	0.82	0.82	0.82	276
weighted avg	0.83	0.83	0.83	276

3.3.1 1 Random forest Classifier Classification Report

5.1.2 Extreme Gradient Boost

The XG Boost performance better than other, here we also use grid search to tuned the hyperparameters and find best values for the parameters.

Grid search found these best combinations:

- n_estimators: 50 (number of boosting rounds)
- max_depth: 5 (maximum depth of each tree)
- learning_rate: 0.1 (shrinkage factor for each tree's contribution)
- subsample: 0.8 (fraction of samples used per tree)
- colsample_bytree: 0.8 (fraction of features used per tree)
- gamma: 2 (minimum loss reduction required to make a split)

With these setting XG Boost achieved:

	PRECISION	RECALL	F1-SCORE	SUPPORT
0 (No Disease)	0.81	0.83	0.82	123
1 (Disease)	0.86	0.84	0.85	153
accuracy			0.84	276
macro avg	0.83	0.84	0.84	276
weighted avg	0.84	0.84	0.84	276

3.3.2 1XG Boost Classifier Classification Report

5.1.3 Support Vector Machine Classifier

The support vector machine is a best choice for binary classification problems, but here their performance is not good if compare to other models. For SVC we also use grid search and after applying grid search we got the following settings:

- C: 1
- gamma: scale
- kernel: “rbf”
- class weight: None (treats both classes equally)

	PRECISION	RECALL	F1-SCORE	SUPPORT
0 (No Disease)	0.79	0.82	0.80	123
1 (Disease)	0.85	0.82	0.84	153
accuracy			0.82	276
macro avg	0.82	0.82	0.82	276

weighted avg	0.82	0.82	0.82	276

3.3.3 1 SVC Classification Report

5.2 ROC Curve

ROC Curve or Receiver Operating Characteristic curve represent binary classifier performance across all possible classification thresholds graphically. ROC Curve plots the TRUE POSITIVE RATE (TPR) against FALSE POSITIVE RATE (FPR). The ROC Curve helps us to find which model distinguish between two classes and it is one of the valuable tool for comparing. If AUC (Area under Curve) close to 1.0 it's mean the model perfectly separates the classes.

If AUC less than 0.5 its mean the model did not separates the classes perfectly.

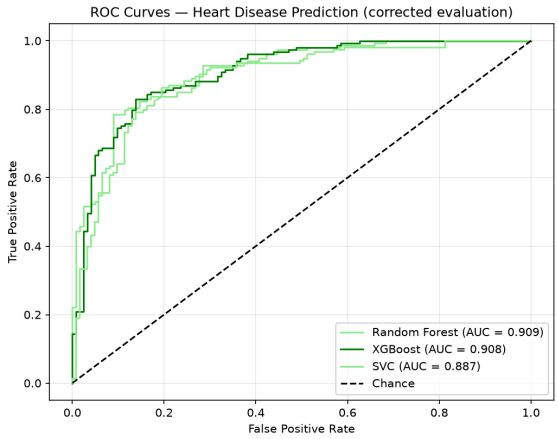


Figure 4.2 1 ROC CURVE

From the graph we can see that XGBoost achieved the highest AUC score of 0.91, closely followed by Random Forest (0.91) and SVC (0.89), which means all three models separate the classes well.

5.3 Discussion of Findings

The performance of each model is outstanding but we will use extreme gradient boost (XG Boost) model to predict the heart disease using user data. The overall accuracy of the models is 82-84% but the AUC score helped us to understand which model was better to separate data in two classes (0 and 1). This means that accuracy is not enough, we need multiple metrics to find the best model. XG Boost perform very well and classified better the data than other models. We select this model for deployment in the website. The user will enter their data using an interactive form and through JavaScript data will be sent to backend for XG Boost to predict the whether the user have heart disease or not. This tool can be easy to use and accessible to everyone. It will useful on those locations where the diagnostic tools like ECG are not available or expensive. This tool can help the doctors or patients to know their heart position and take steps early to change their lifestyle or medical treatment.

Chapter 6. Conclusions

In our final year project, we developed a web-based heart disease prediction using machine learning which gives instant results to the user according to their data. In this study, we trained three machine learning models Random Forest Classifier, XG Boost and Support Vector Machine Classifier. All models perform well on the UCI dataset but the XG Boost model AUC score is higher than the others which indicates that this model perfectly separates the classes: Heart Disease (1) and No Heart Disease (0).

6.1 Key Findings

Our key findings in this study are:

6.1.1 Model Performance

Based on the overall accuracy and the ROC Curve, extreme gradient boost with 0.91 AUC score and 84% overall accuracy outperforms both Random Forest and Support Vector Classifier. We deploy XG Boost model on a website.

6.1.2 Web Application

Based on AUC score and classification report we choose XG Boost to our main model. We integrated XG Boost model into a flask-based simple web app so user can enter their data like age, cp, cholesterol, blood pressure and other features easily.

6.2 Contributions

6.2.1 Practical Tool

Our main goal was to develop a practical tool that can be easy to use and accessible to everyone without any specialized software and equipment. We integrate the best performer model into a simple and easy to use website. The user even non-experts

(patients or clinicians) can use this tool easily. The backend flask handle user data like data cleaning, convert categorical to numerical, scaling data.

6.3 Limitations

6.3.1 Data Scope

To train our models, we used only one UCI heart disease dataset which contains only 920 patient records. The result may be different on other or large datasets. This dataset only contain 13 clinical features other important factors like ECG waveforms or lifestyle habits were not included.

6.3.2 Prototype Design

The website is easy to use but it's only a demonstration. There are no advanced features like user accounts, data storage and compliance with medical data standards.

6.4 Recommendation for Future Work

This study successfully develop a working prototype for heart disease prediction using machine learning. But there are several areas for research and development to improve and expand the system.

6.4.1 Model Maintenance

The machine learning models are very powerful and it can find hidden patterns that impossible for human. But models can be outdated if they are not updated with new data regularly. Over time the patterns in patient data may change due to lifestyle and environment changes. Therefore retrain the machine learning model with new data from hospital databases or research databases. Use the automated pipelines that updated the model with new available data.

6.4.2 Feature Expansion

The current model use only 13 features which available in UCI Heart Disease Dataset. There are many other factors that can affect heart risk. If we train model with more health indicators the model can offer better predictions.

6.4.2.1 Example of Other Features

- ECG (Electrocardiogram) data: If we add waveform data from heart tests it can help to detect abnormalities that regular numbers cannot capture.
- Lifestyle data: We can include diet habits, exercise frequency, stress level and sleep pattern.
- Family History & Genetics: Some people are higher heart risk due to their close relative heart problems
- Wearable Devices: Like smartwatches we can integrate data (heart rate, physical activity) to get real time risk monitoring.

Adding these features can improve our model more accurate and personalized prediction.

6.4.3 Clinical Integration

If we designed system with healthcare professionals in mind, it can be usable tool in real hospitals and clinics. Some Suggestion to improve the system:

- Secure User Authentication: Doctors and patients should have private and protected accounts.
- Data Storage and Encryption: We can store patient data securely using standard encryption techniques like AES or RSA.

- EHR Integration: To connect with existing Electronic Health Record (EHR) systems through which doctors can directly access and update the patient risk score.

6.4.4 Mobile App Deployment

Most people use smartphones and they have access to mobile anytime. Creating a mobile application version to improve accessibility. Some suggestion includes:

- Develop mobile app for android and IOS using platforms like React Native or Flutter.
- Allow user to enter their data and instantly get output.
- If we include different languages it will make more user-friendly.

References

[1] "Cardiovascular diseases," [Online]. Available: https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_1.

[2] "Cardiovascular diseases," [Online]. Available: https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_1.

[3] "The Lancet Global Health," [Online]. Available: https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(22)00497-1/fulltext. [Accessed 2019].

[4] "UC Irvine Machine Learning Repository," 30 6 1988. [Online]. Available: https://archive.ics.uci.edu/dataset/45/heart+disease.

[5] "Cleveland Clinic," [Online]. Available: https://my.clevelandclinic.org/health/diseases/21493-cardiovascular-disease.

[6] Y. e. al., "International Study Highlights 9 Controllable Risk Factors Responsible for 90% of Heart Attacks Worldwide," 13 October 2004. [Online]. Available: https://www.audiologyonline.com/releases/international-study-highlights-9-controllable-5393?utm_source=chatgpt.com.

[7] H. F. El-Sofany, "Predicting Heart Diseases Using Machine Learning and Different Data Classification Techniques," *IEEE,* vol. 12, pp. 106146-106160, 2024.

[8] M. S. G. A. a. J. R. N. L. Fitriyani, "HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System," *IEEE,* vol. 8, pp. 133034-133050, 2020.

[9] E. A. A. S. B. a. S. A. A. A. Almazroi, "A Clinical Decision Support System for Heart Disease Prediction Using Deep Learning," *IEEE,* vol. 11, pp. 61646-61659, 2023.

[10] S. A. M. A. Z. A. S. H. A. a. A. S. M. S. A. Reshan, "A Robust Heart Disease Prediction System Using Hybrid Deep Neural Networks," *IEEE,* vol. 11, pp. 121574-121591, 2023.

[11] A. R. K. M. a. M. S. A. A. M. Qadri, "Effective Feature Engineering Technique for Heart Disease Prediction With Machine Learning," *IEEE,* vol. 11, pp. 56214-56224, 2023.

[12] H. F. S. U. A. S. S. a. I. G. N. Ahmad, "Efficient Medical Diagnosis of Human Heart Diseases Using Machine Learning Techniques With and Without GridSearchCV," *IEEE,* vol. 10, pp. 80151-80173, 2022.

[13] T. Z. a. D. A. C. F. Shamout, "Machine Learning for Clinical Outcome Prediction," *IEEE ,* vol. 14, pp. 116-126, 2021.

[14] S. D. Avijit Kumar Chaudhuri, "The Performance of Feature Selection Approaches on Boosted Random Forest Algorithms for Predicting Cardiovascular Disease," *Taylor & Francis Group,* no. https://www.taylorfrancis.com/chapters/edit/10.1201/9781003429609-18/performance-feature-selection-approaches-boosted-random-forest-algorithms-predicting-cardiovascular-disease-avijit-kumar-chaudhuri-sulekha-das, p. 23, 2024.

[15] A. J. W. B. W. S. S. C. a. P. M. Asim Saleem, "Neural-XG Boost: A Hybrid Approach for Disaster Prediction and Management Using Machine Learning," *IEEE Access,* vol. 13, pp. 86768-86780, 2025.